

Retail Sales & Business Performance Analytics

BRD • FRD • Case Study • Power BI Dashboard

1. Business Requirements Document (BRD)

This BRD defines the business problem, objectives, stakeholders, scope, requirements and proposed reporting solution.

1.1 Business Problem

Management lacks a consolidated view of retail sales performance. Data comes from multiple sources and requires manual consolidation. The dataset contains duplicates, missing values, inconsistent entries, and invalid or suspicious values, making monthly reporting slower and comparisons difficult.

1.2 Business Objectives

- Create a reliable cleaned dataset for reporting.
- Monitor monthly sales and order performance.
- Analyze product, category, salesperson, region, city, and channel performance.
- Monitor completed, returned, and cancelled orders.
- Analyze customer and payment-method patterns.
- Evaluate discount and sales performance.
- Provide an interactive Power BI dashboard.

1.3 Stakeholders

Stakeholder	Use
Senior Management	Executive sales and performance overview
Sales Manager	Product, salesperson, region, and channel performance
Regional Managers	Regional and city performance
Finance	Sales, discounts, returns, cancellations, payment methods
Operations	Order status and data-quality monitoring
Marketing / CRM	Customer and channel insights
IT / Data Team	Data quality and reporting support
Business / Data Analyst	Requirements, validation, analysis, dashboard and insights

1.4 Scope

- Sales, orders, products, customers, regions/cities, channels, payment methods, order status, and salesperson analysis.
- Data profiling, cleaning, validation, Excel analysis, SQL analysis, and Power BI reporting.
- Interactive filtering and management-level reporting.

1.5 Business Requirements

ID	Requirement
BR-01	Data Accuracy – identify/remove exact duplicate records and validate key fields.
BR-02	Missing Data Management – identify missing values and handle them appropriately.
BR-03	Monthly Sales Monitoring – track sales trends by month.
BR-04	Sales Performance – analyze product, category, salesperson, region, city, and channel.
BR-05	Order Status Monitoring – monitor completed, cancelled, and returned orders.
BR-06	Customer Analysis – analyze customer activity and order patterns.
BR-07	Regional Performance – compare sales and orders across regions.
BR-08	Channel Performance – compare Online, Store, and Marketplace performance.
BR-09	Discount Analysis – evaluate discounts and sales.
BR-10	Data Quality Validation – identify invalid quantities, discounts, dates, and inconsistent values.
BR-11	Interactive Reporting – provide slicers, KPIs, charts, and tables in Power BI.

1.6 Proposed Solution

Raw Sales Data → Data Profiling & Quality Checks → SQL Cleaning & Validation → Cleaned Dataset → Excel Analysis & Validation → Power BI Data Model + DAX → Interactive Dashboard → Business Insights

2. Functional Requirements Document (FRD)

The FRD translates business requirements into functional behavior for data processing, analysis, and Power BI reporting.

2.1 Data Structure

Area	Fields
Identifiers	Order_ID, Customer_ID, Product_ID
Date	Order_Date
Product	Product_Name, Category
Geography	Region, City
Measures	Quantity, Unit_Price, Discount, Sales_Amount
Operations	Sales_Channel, Payment_Method, Order_Status, Salesperson
Cleaning Helpers	Order_Status_Clean, Sales_Channel_Clean, Payment_Method_Clean, Category_Clean, Quantity_Status, Quantity_Review, Discount_Status, Gross_Sales, Net_Sales

2.2 Functional Requirements

ID	Function	Expected Behavior
FR-01	Data Import	Import Clean_Data from saved Excel workbook.
FR-02	Data Types	Set dates, quantities, prices, discounts, and sales correctly.
FR-03	Date Dimension	Create DateTable for time reporting.
FR-04	Relationship	Relate DateTable[Date] to Clean_Data[Order_Date].
FR-05	Total Sales	Calculate completed-order sales using Net_Sales.
FR-06	Total Orders	Calculate distinct Order_ID count.
FR-07	Total Quantity	Calculate total quantity.
FR-08	Average Order Value	Calculate Total Sales / Total Orders.
FR-09	Status Metrics	Calculate completed, returned, and cancelled orders.
FR-10	Date Filtering	Filter by reporting period.
FR-11	Business Filters	Filter by Region, Category, Channel, and Status.
FR-12	Trend Analysis	Display monthly sales trend.
FR-13	Product Analysis	Display top products by sales.
FR-14	Category Analysis	Display sales by category.
FR-15	Regional Analysis	Display sales by region.
FR-16	Performance Table	Display category sales, orders, and average order value.

2.3 Key DAX Measures

```
Total Sales =  
CALCULATE(SUM(Clean_Data[Net_Sales]),  
    Clean_Data[Order_Status_Clean] = "Completed")
```

```
Total Orders =  
DISTINCTCOUNT(Clean_Data[Order_ID])
```

```
Total Quantity =  
SUM(Clean_Data[Quantity])
```

```
Average Order Value =  
DIVIDE([Total Sales], [Total Orders], 0)
```

```
Cancelled Orders =  
CALCULATE(DISTINCTCOUNT(Clean_Data[Order_ID]),  
    Clean_Data[Order_Status_Clean] = "Cancelled")
```

```
Returned Orders =  
CALCULATE(DISTINCTCOUNT(Clean_Data[Order_ID]),  
    Clean_Data[Order_Status_Clean] = "Returned")
```

2.4 Dashboard Components

Component	Purpose
Date slicer	Filter reporting period
Region slicer	Filter regional performance
Category slicer	Filter categories
Sales Channel slicer	Compare Online, Store, Marketplace
Order Status slicer	Filter Completed, Cancelled, Returned
KPI Cards	Show Total Sales, Orders, Quantity, AOV
Monthly Sales Trend	Show sales movement over time
Sales by Product	Identify products contributing to sales
Sales by Category	Show category contribution
Sales by Region	Compare regional sales
Category Performance	Show category sales, orders, AOV

3. Business Analytics Case Study

3.1 Project Overview

A fictional retail and e-commerce company used for this portfolio case study. The objective is to transform a messy retail transaction dataset into a validated analytical dataset and interactive Power BI dashboard.

3.2 Dataset

Attribute	Details
Dataset Type	Synthetic but intentionally realistic and messy retail/e-commerce data
Cleaned Records	1,505
Original Fields	16
Cleaned/Helper Fields	25 total columns
Channels	Online, Store, Marketplace
Categories	Accessories, Electronics, Furniture, Stationery
Statuses	Completed, Cancelled, Returned

3.3 Data Problems Identified

- Exact duplicate rows.
- Repeated Order_ID values requiring careful interpretation.
- Missing Customer_ID, City, Payment_Method, Salesperson, Quantity, and Discount.
- Zero, negative, and unusually high quantities.
- Discount values outside the expected range.
- Invalid/non-date values in Order_Date requiring validation before Power BI date conversion.

3.4 Analysis Workflow

Stage	Work Performed	Output
1. Requirements	Defined business questions, stakeholders, scope, KPIs	BRD / FRD
2. Data Profiling	Reviewed fields, missing values, duplicates, invalid values	Data-quality findings
3. SQL	Filtering, aggregation, validation, grouping, business analysis	Analytical queries
4. Excel	Raw_Data, Clean_Data, Analysis workflow	Cleaned dataset and analysis
5. Power BI	Data model, DAX, slicers, charts, tables	Interactive dashboard
6. Validation	Compared Power BI results with Excel	Validated metrics

3.5 Dashboard Results

Metric	Displayed Value
Total Sales	29.59M
Total Orders	2K
Total Quantity	5K
Average Order Value	19.73K

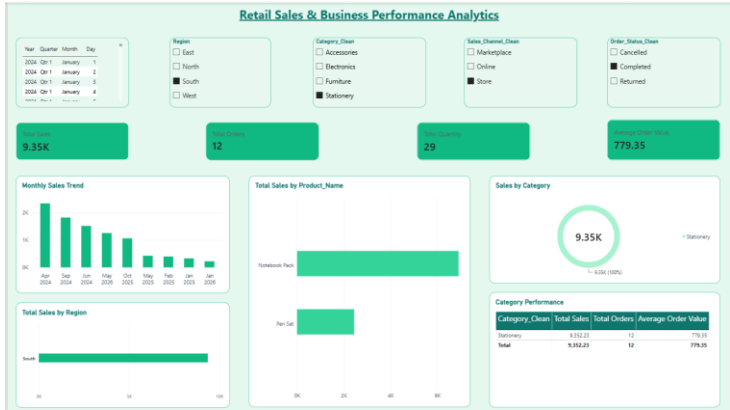
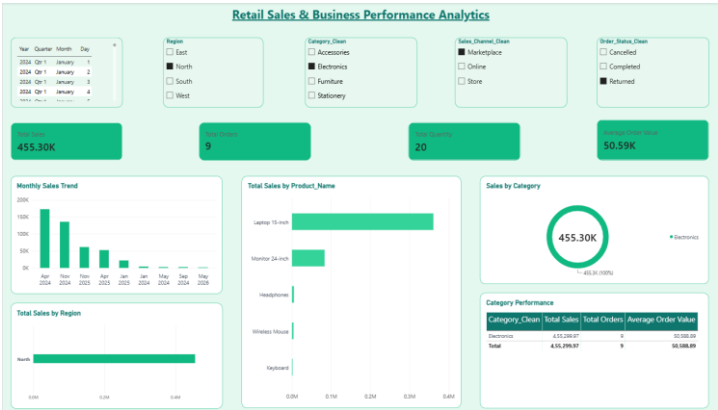
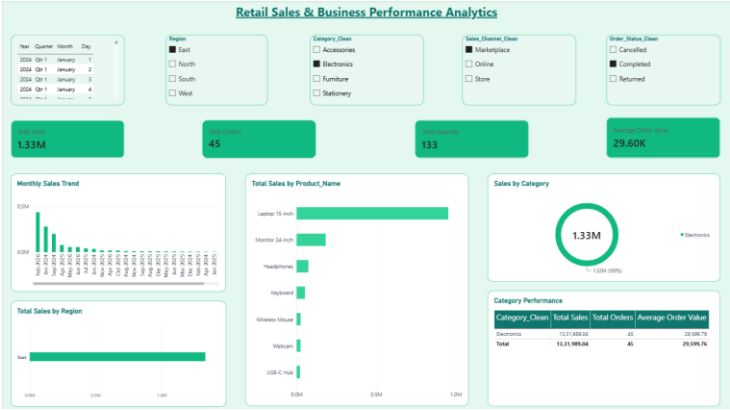
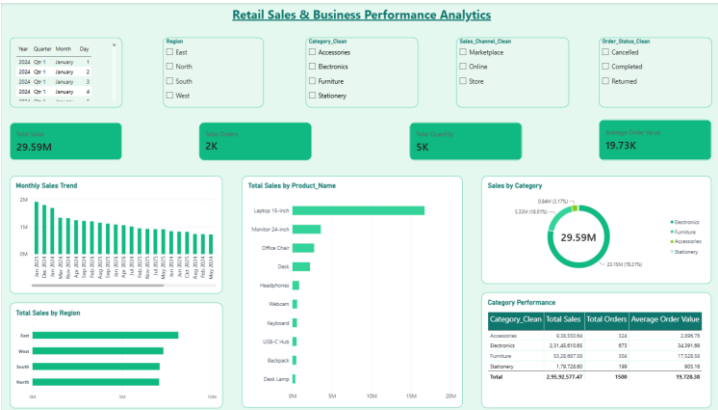
3.6 Business Insights Available

- Monthly trend highlights high- and low-performing periods.
- Product analysis highlights major contributors to sales.
- Category analysis shows contribution by product category.
- Regional analysis compares East, West, South, and North.
- Order-status filtering isolates completed, returned, and cancelled transactions.
- Multiple slicers allow focused analysis by date, region, category, channel, and status.

3.7 Business Value

- Centralized view of sales performance.
- Interactive business comparisons.
- Improved visibility into data-quality issues.
- Repeatable workflow from raw data to dashboard.
- Faster identification of products, categories, regions, and periods requiring attention.

4. Final Power BI Dashboard



5. Project Deliverables

Deliverable	Description
BRD	Business problem, objectives, stakeholders, scope, and business requirements
FRD	Functional requirements, data rules, DAX measures, dashboard functions
Case Study	End-to-end analytics workflow, data issues, results, and business value
Excel	Raw_Data, Clean_Data, and Analysis workflow
SQL	Data validation and business analysis
Power BI	Interactive Executive Overview dashboard

6. Tools Used

- Microsoft Excel – cleaning, validation, calculations, analysis.
- MySQL – querying and business analysis.
- Microsoft Power BI – data modeling, DAX, visualization, reporting.
- ChatGPT – analytical assistance, documentation, DAX.